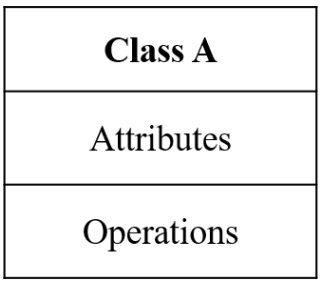
A) Develop class diagram to represent data/class design model

Class diagrams are used to represent class-based model category of requirement model.

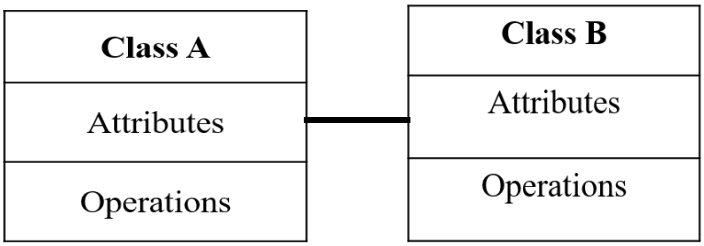
A class diagram provides a **static or structural view of a system**.

UML class diagram model includes classes with

- class name,
- their attributes,
- operations,



Class diagram also represents relationships between classes.



Steps to draw class diagram

1. Identifying Classes-

Classes are determined by **identifying each noun or noun phrase** in problem statement.

Class category	Examples
External entities	Other systems, devices, people
Things	Reports, Displays, Letters, Signals
Occurrences or events	Movement of elevator, robot
Roles	Manager, Engineer, Salesperson, Student
Organizational units	Class, Division, Group, Project-team
Places	Manufacturing floor, PC assembly workshop, delivery section
Structures	Sensors, vehicles, or computers

2. Specifying Attributes

Attributes **describe a class** that has been selected for inclusion in the requirements model.

Examples:

Library Member- Name, Dept, Contact_No, MemberID

Account – Act_Number, Type of account, Balance

Product - Prd_ID, Prd_Name, Prd_Category

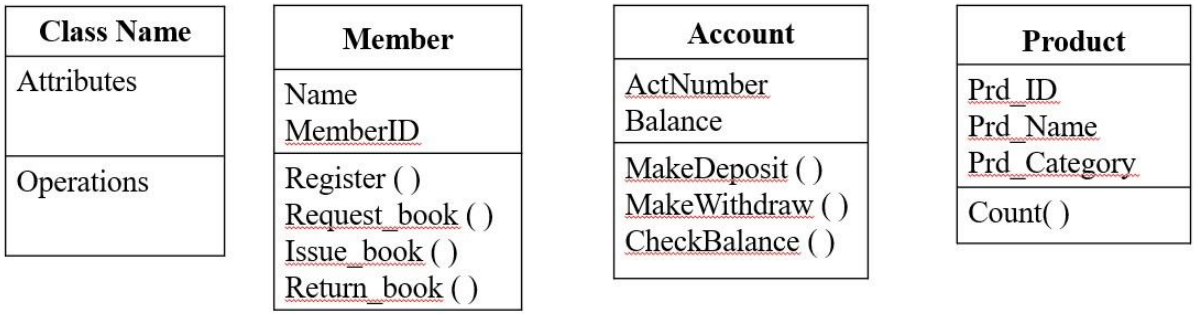
3. Defining Operations

Operations **define the behavior of an object**.

Four broad categories of operations are

Operation categories	Examples
Manipulate data	Adding, Deleting, Selecting
Perform a computation	Calculate total, Count number of items in order
Inquire about the state of an object	Verify issue count, Register member
Monitor an object for the occurrence of a controlling event	Alarm, Signal generation

Examples- representation of class diagram



Class in Library management system

Class in Bank management system

Class in Order management system

Class diagram with relationships between classes

In class diagram, classes are **interrelated to each other** in specific ways.

These relationships in class diagrams include different types of **logical connections**.

The **different types** of logical connections/relationships that are possible in UML are,

1. **association**

2. **multiplicity**
3. **generalization (inheritance)**
4. **aggregation**
5. **composition**

1. Association

It is a broad term that encompasses **any logical connection or relationship between classes**.

For example, classes library member and book may be associated as :



Classes person and car may be linked as :



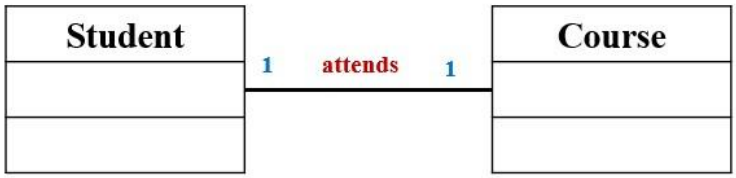
2. Multiplicity

The multiplicity of one end of an association means the **number of objects of that class associated with the other class**.

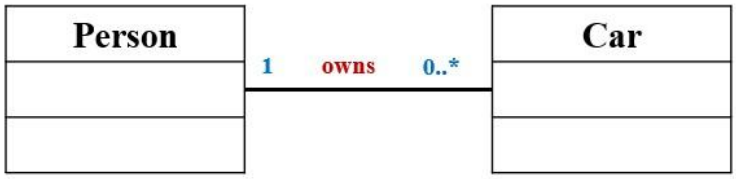
A multiplicity is specified **by a non-negative integer** or by a **range of integers**.

Multiplicity	Meaning
1	Exactly one
0..1	Zero or one
1..*	One or more
0..* or *	Zero or more

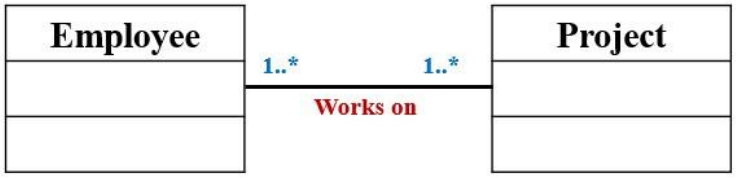
A student can register for one course only.



A person owns zero or more cars.



Many employees work on more than one project.

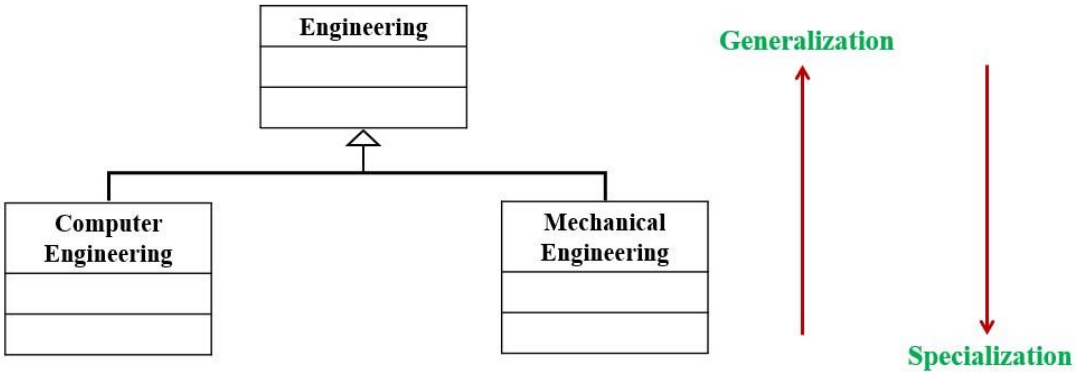


3. Generalization (inheritance)

A generalization is a relationship between a **parent class** (superclass) and a **child class** (subclass).

In this, the child class is **inherited from** the parent class.

Example -

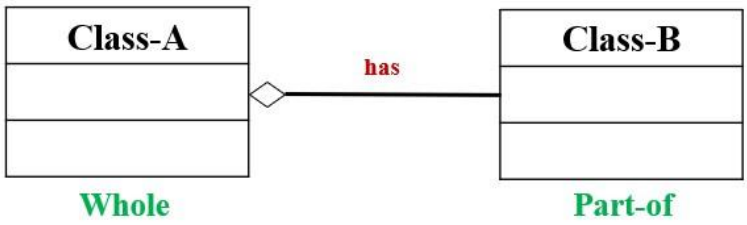


4. Aggregation

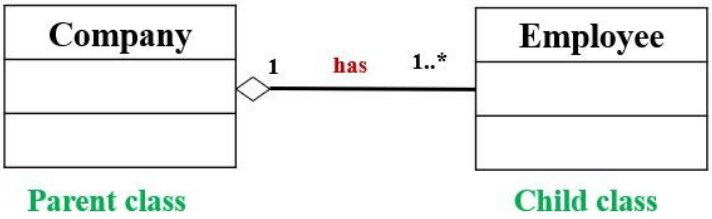
An aggregation is a **special kind of association**, which represents **has a relationship**.

It defines a **whole/part** or **part-of** relationship.

In this kind of relationship, the **child class can exist independently of its parent class**. For example, Class-B is a part of the Class-A (class B exist independently of class A),



For example, the class "Company" has one or more employees (Employee is part of Company)



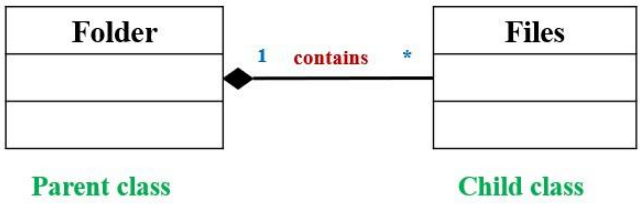
Classes Employee may exist independently of their parent classes.

5. Composition

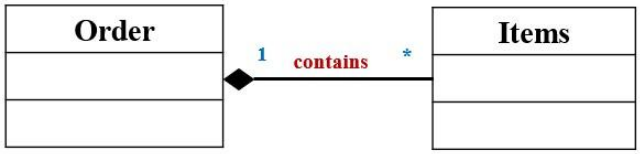
A composition is an aggregation (**whole/part** or **part-of** relationship) indicating **strong ownership** of the parts.

In a composition, the **parts (child class) live and die with the owner (parent class)** because they have no role in the software system independent of the owner.

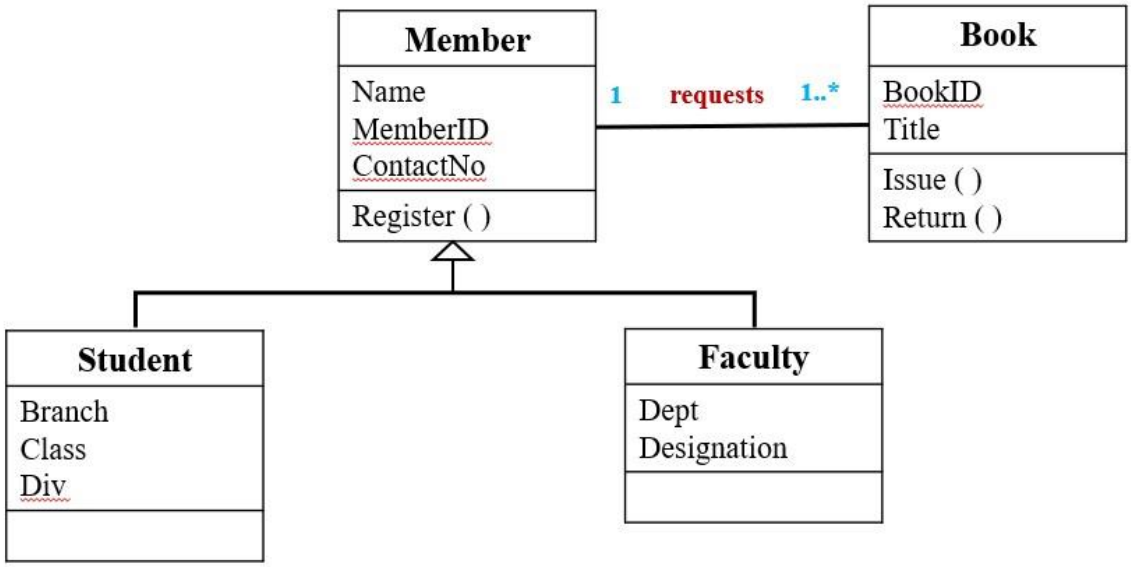
For example, the class "Folder in computer drive" is composed of class "Files" (files get deleted with folder).



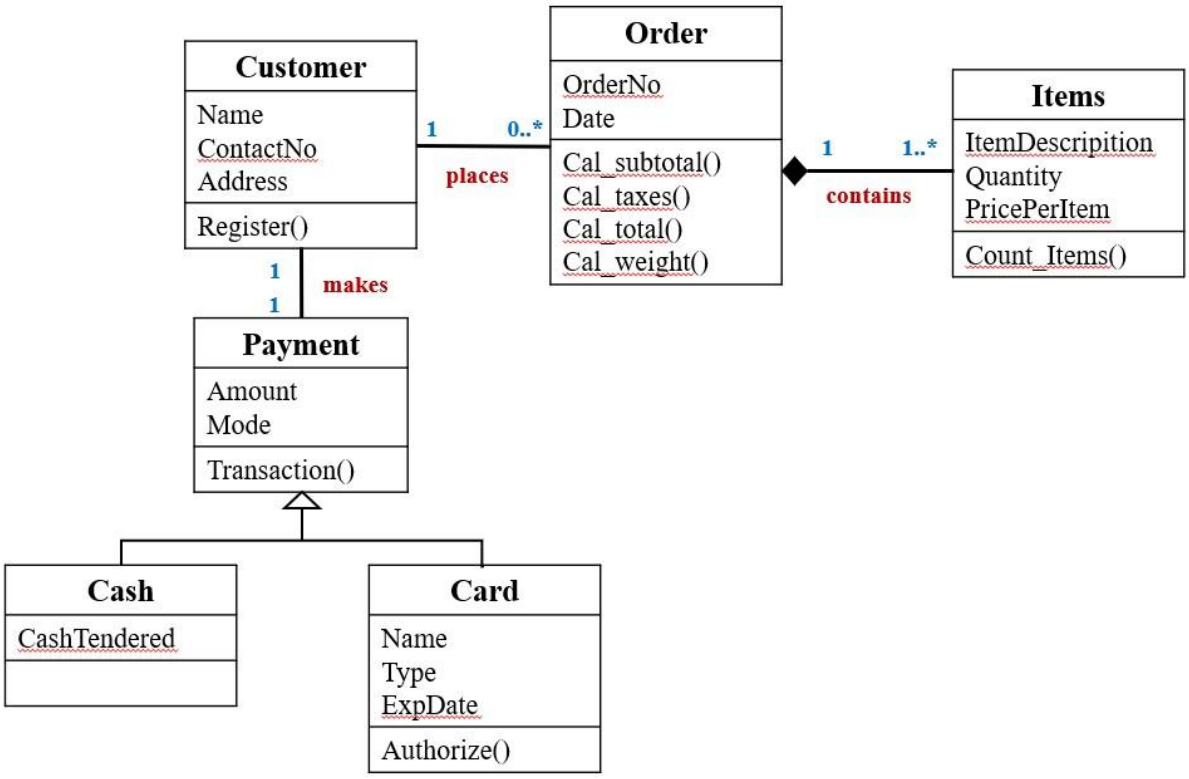
Another example can be where Items will be removed from list if Order is cancelled,



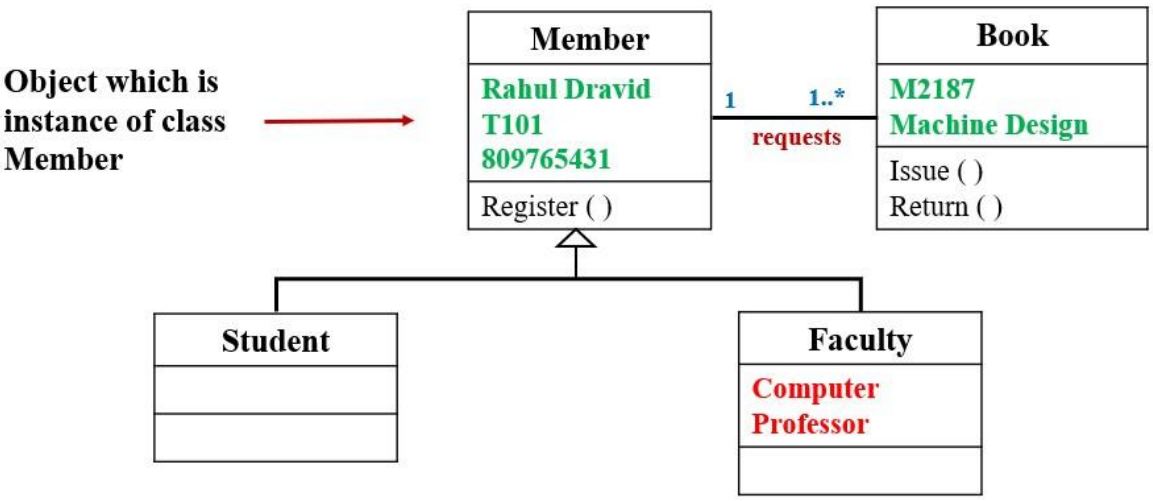
Class diagram for Library Management System



Class diagram for Order Management System



Object diagram for Library Management System



https://www.javatpoint.com/uml-class-diagram

B) Develop State chart Diagrams

A Statechart diagram describes different states of a component in a system. The states are specific to a component/object of a system.

Activity diagram is a special kind of a Statechart diagram. As Statechart diagram defines the states, it is used to model the lifetime of an object.

Purpose of Statechart Diagrams

Statechart diagram is used to model the dynamic nature of a system. They define different states of an object during its lifetime and these states are changed by events. Statechart diagrams are useful to model the reactive systems. Reactive systems can be defined as a system that responds to external or internal events.

Statechart diagram describes the flow of control from one state to another state. **States are defined as a condition in which an object exists and it changes when some event is triggered.** The most important purpose of Statechart diagram is to model lifetime of an object from creation to termination.

How to Draw a Statechart Diagram?

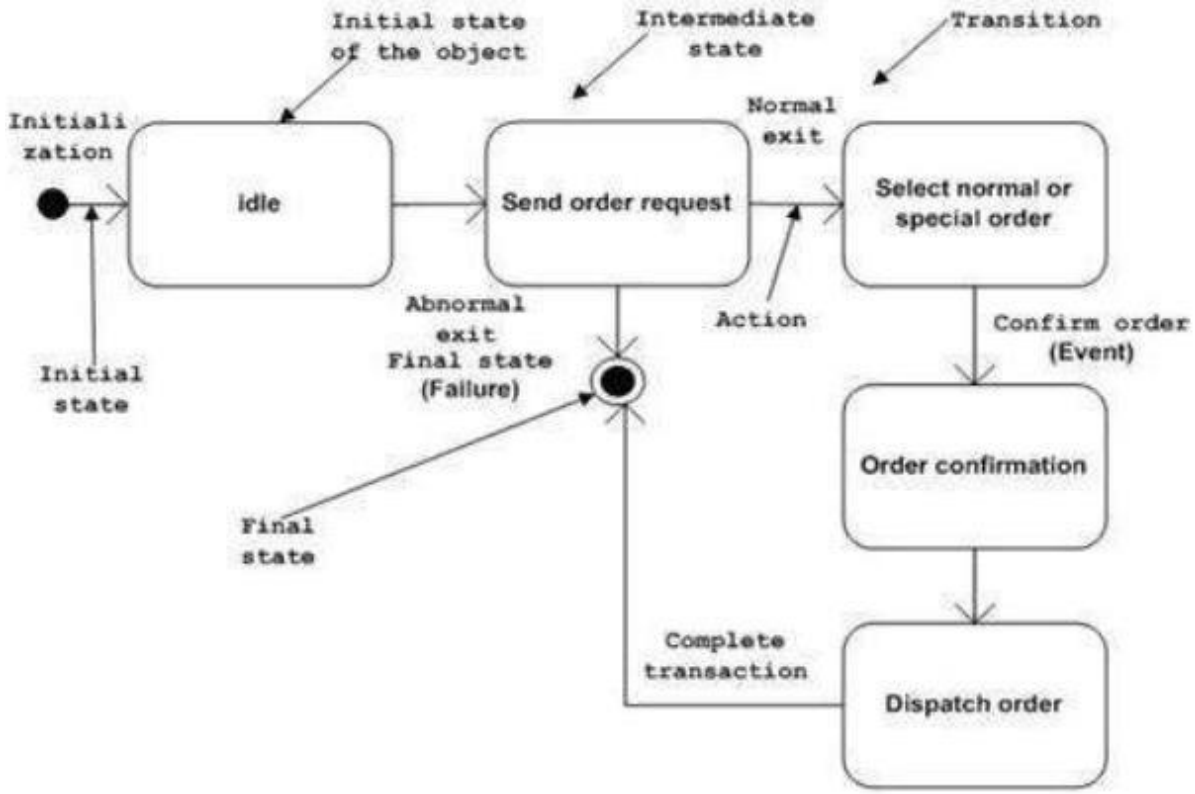
Before drawing a Statechart diagram we should clarify the following points −

- Identify the important objects to be analyzed.
- Identify the states.
- Identify the events.

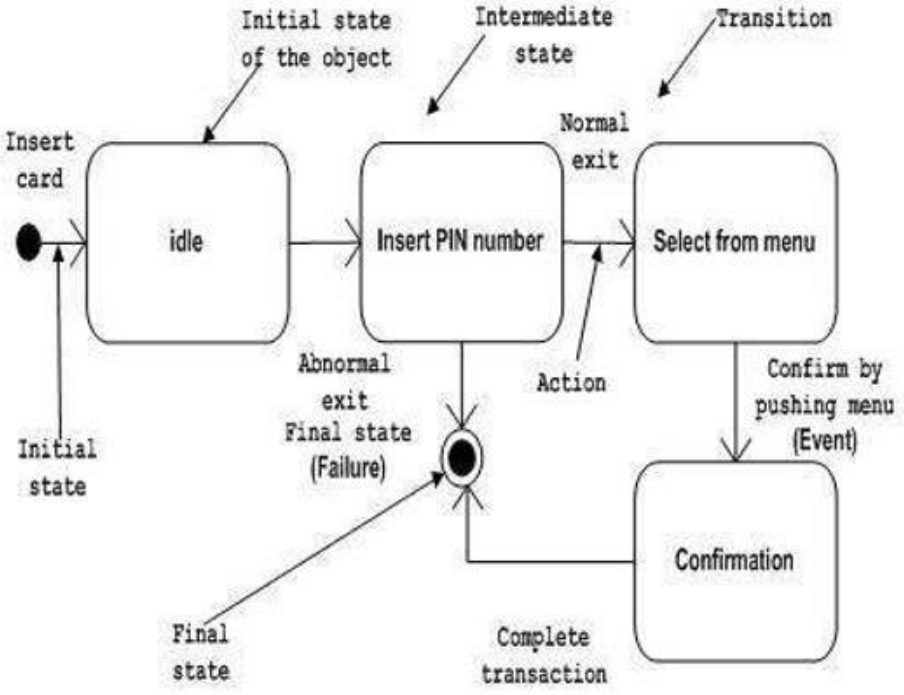
Following is an example of a **Statechart diagram for order management system**. Here the state of Order object is analyzed

The first state is an **idle state from where the process starts**. The next states are arrived for events like send request, confirm request, and dispatch order. These events are responsible for the state changes of order object.

During the life cycle of an object (here order object) it goes through the following states and there may be some abnormal exits. This abnormal exit may occur due to some problem in the system. When the entire life cycle is complete, it is considered as a complete transaction as shown in the following figure. The initial and final state of an object is also shown in the following figure.



Statechart diagram for money withdrawal from ATM-



https://www.tutorialspoint.com/uml/uml_statechart_diagram.htm